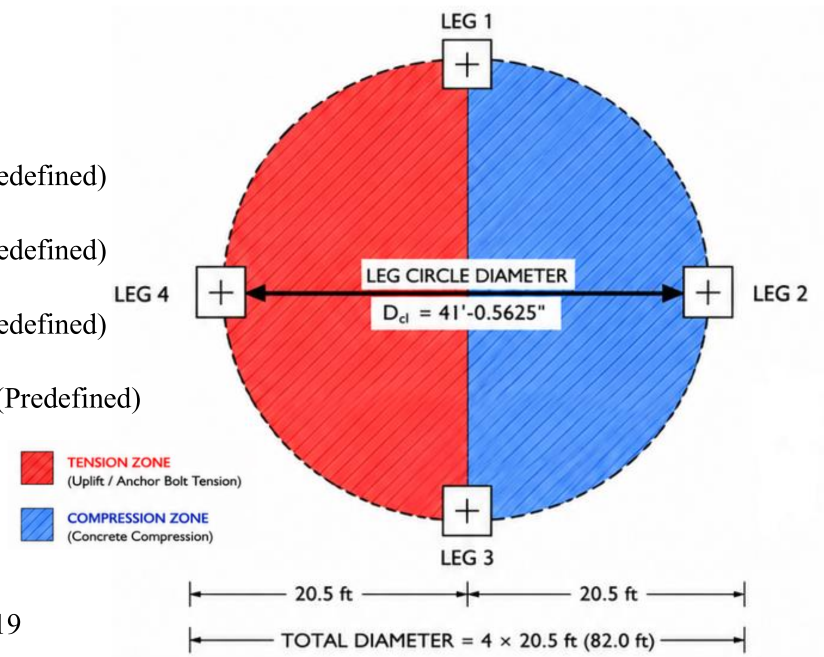
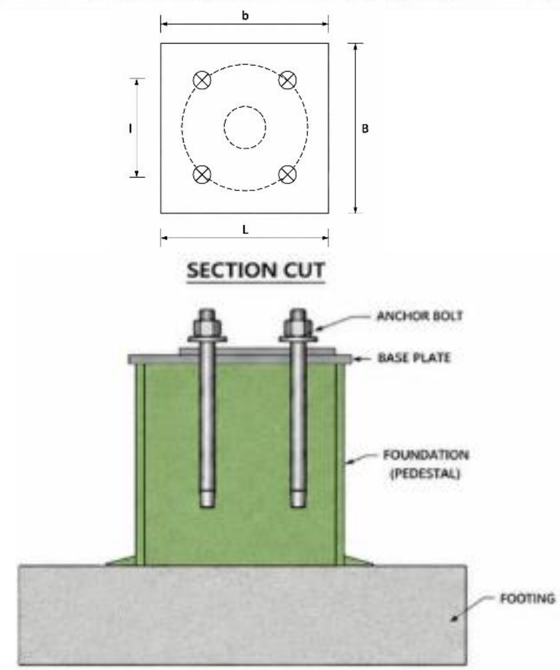
3: Foundation Design:

Design Parameter	Symbol	Value	Source
Number of legs	N_{leg}	**4**	User Input
Total factored axial load	P_u	**1,609.756 kips**	Main Tank Design (Predefined)
Governing factored overturning moment	M_u	**5,673.437 kip-ft**	Main Tank Design (Predefined)
Total factored horizontal shear	V_u	**52.30 kips**	Main Tank Design (Predefined)
Leg circle diameter	D_{cl}	**41'-0.5625"**	Main Tank Geometry (Predefined)
Pedestal size	$B \times L$	**39 in × 39 in**	User Input
Base plate size	$b_{bp} \times L_{bp}$	**30 in × 30 in**	User Input
Concrete compressive strength	f_c'	**4.0 ksi**	User Input / ACI 318-19
Reinforcing steel yield strength	f_y	**60 ksi**	User Input / ACI 318-19
Anchor bolts	$N_b \times d_b$	**4 × 1.75 in**	See Segment 1 – Anchor Bolts
Anchor embedment depth	h_{ef}	**40 in**	User Input
Pedestal clear height used	H_p	**≈ 7'-5"**	User Input
Concrete unit weight	γ_c	**150 pcf**	User Input





Step 1 — Gravity axial load per leg

For four equally spaced legs:

$$P_{gravity} = \frac{P_u}{N_{leg}}$$
$$P_{gravity} = \frac{1609.756}{4}$$
$$\boldsymbol{P_{gravity} = 402.44\ \textbf{kip/leg}}$$

Step 2 — Overturning force

From the anchor-bolt design:

$$r = 246.28\ \text{in}$$

and the overturning force calculation gives:

$$T = 207.3\ \text{kip}$$

with two legs on the tension side.

Therefore:

$$T_{leg} = \frac{207.3}{2}$$
$$\boldsymbol{T_{leg} = 103.65\ \textbf{kip}}$$

This is the same tension-leg reaction established in the anchor-bolt design.

For pedestal design, use this force conservatively as the uplift/tension transfer demand.

Maximum Compression Pedestal Load

The compression-side leg receives the gravity reaction plus the overturning reaction:

$$P_{comp} = Pu_{,leg} + T_{leg}$$

$$P_{u,ped} = 402.44 + 103.65$$

$$\mathbf{P_{u,ped} = 506.09\ kip}$$

The pedestal self-weight can also be included.

For approximately 7'-5" pedestal height:

γ_c=150 pcf

$$W_p = (3.25)(3.25)(7.417)(0.150)$$
$$W_p \approx 11.75\ \text{kip}$$

Factored:

$$1.2W_p = 14.10\ \text{kip}$$

Therefore, conservatively:

$$P_{comp,total} = 506.09 + 14.10$$
$$\boldsymbol{P_{comp,total} \approx 520.2\ \mathbf{kip}}$$

Step 3 - Pedestal Concrete Bearing Check

Base plate:

$$A_1 = \frac{\pi(30)^2}{4} = 706.86\ \text{in}^2$$

Pedestal:

$$A_2 = 39(39) = 1521\ \text{in}^2$$

Bearing enhancement factor:

$$\sqrt{\frac{A_2}{A_1}} = \sqrt{\frac{1521}{706.86}} = 1.467$$

Using the concrete bearing-strength relationship: $\phi B_n = \phi(0.85 f_c' A_1)\sqrt{\frac{A_2}{A_1}}$

Taking:

$$\phi = 0.65$$
$$\phi P_n = 0.65(0.85)(4)(706.86)(1.30)$$
$$\boldsymbol{\phi P_n \approx 2292} \textbf{ kips}$$

Demand:

$$P_u = 520.2 \text{ kip}$$
$$\text{D/C} = \frac{520.2}{2292}$$

Therefore:

0.23 < 1.00…….**CONCRETE BEARING — PASS**

Demand/capacity ratio:…………………………………………………… $D/C \approx 0.20$

So the **39 × 39 in pedestal is more than adequate for bearing**.

Step 4 - Pedestal longitudinal Reinforcement

The approved foundation drawing uses:

18 − #6 vertical bars

Area of one #6:

$$A_b = 0.44 \text{ in}^2$$

Therefore:

$$A_s = 18(0.44)$$
$$\boldsymbol{A_s = 7.92 \text{ in}^2}$$

$$A_g = 39(39)$$

$$A_s = 1521\ in^2$$

Reinforcement ratio:

$$\rho = \frac{7.92}{39(39)}$$
$$\rho = 0.0053$$

or:…………………………………………………………………………$\boldsymbol{\rho = 0.53}\%$

For the cast-in-place pedestal-to-foundation interface, ACI 318-19 requires reinforcement crossing the interface of at least:

$$A_{s,min} = 0.005A_g$$

ACI 318-19 Section 16.3.4.1 specifies the $0.005A_g$minimum for connections between a cast-in-place column or pedestal and its foundation.

As,min=0.001(1521)

$$A_{s,min} = 7.605\ in^2$$

Provided:

$$A_s = 7.92 \text{ in}^2$$

Therefore:

$$7.92 > 7.605$$

$$\boxed{\textbf{PASS}}$$

Reinforcement ratio:

$$\rho_g = \frac{7.92}{1{,}521}$$
$$\rho_g = 0.00521 = 0.521\%$$

Step 5 - Axial Compression Capacity

Gross area:

$$A_g = 1{,}521 \text{ in}^2$$

Steel area:

$$A_s = 7.92 \text{ in}^2$$

Nominal concentric compression strength:

$$P_o = 0.85 f_c' \left(A_g - A_s\right) + f_y A_s$$
$$P_o = 0.85(4.0)(1{,}521 - 7.92) + 60(7.92)$$
$$\boldsymbol{P_o = 5{,}619.7 \textbf{ kips}}$$

Using a conservative tied-member design strength factor and 0.80 maximum axial-strength factor:

$$\phi P_n = 0.65(0.80)(5{,}619.7)$$
$$\boldsymbol{\phi P_n = 2{,}922.2 \textbf{ kips}}$$

Demand:

$$P_u = 520.2 \text{ kips}$$

Therefore:

$$\text{D/C} = \frac{520.2}{2{,}922.2}$$
$$\textbf{D/C} = \boldsymbol{0.18}$$

Step 6 - Pedestal Uplift Check

Governing tension-side pedestal demand:

$$\boldsymbol{T_u = 103.65\ \mathrm{kips}}$$

Provided longitudinal reinforcement:

$$A_s = 7.92\ \mathrm{in}^2$$

Using:

$$\phi = 0.90$$

and:

$$f_y = 60\ \mathrm{ksi}$$

The available steel tension strength is:

$$\phi T_n = 0.90(7.92)(60)$$
$$\boldsymbol{\phi T_n = 427.68\ \mathrm{kips}}$$

Demand/capacity ratio:

$$\mathrm{D/C} = \frac{103.65}{427.68}$$
$$\mathbf{D/C = 0.24}$$
$$\boxed{\mathrm{PASS}}$$

Therefore, the 18-#6 vertical reinforcement has adequate theoretical tensile capacity for the pedestal uplift-transfer demand.

Step 7 - Horizontal Shear per Pedestal

Total tank shear:

$$V_u = 52.30 \text{ kip}$$

Assuming equal distribution among four legs:

$$V_{u,ped} = \frac{52.30}{4}$$
$$\boldsymbol{V_{u,ped} = 13.075 \text{ kip}}$$

Thus:

$$\boldsymbol{V_{u,ped} = 13.08 \text{ kips}}$$

This is consistent with the anchor design, which subsequently distributes the pedestal shear among four anchor bolts and obtains 3.27 kip/bolt.

Step 8 – Pedestal one-Shear Check

Use:

$$d \approx 35.4 \text{ in}$$

and:

$$b_w = 39 \text{ in}$$

For a conservative concrete shear estimate:

$$V_c = 2\sqrt{f_c'}\, b_w d$$

with:

$$f_c' = 4{,}000 \text{ psi}$$

gives approximately:

$$\boldsymbol{V_c \approx 174.5 \text{ kips}}$$

Using:

$$\phi = 0.75$$
$$\phi V_c = 0.75(174.5)$$
$$\boldsymbol{\phi V_c \approx 130.9\ \textbf{kips}}$$

Demand:

$$V_u = 13.075\ \text{kips}$$

Therefore:

$$\text{D/C} = \frac{13.075}{130.9}$$
$$\textbf{D/C} \approx \mathbf{0.10}$$

Step 9 – Pedestal Flexure Check

Conservatively model the pedestal as a cantilever from the top of the footing.

$$M_u = V_u H_p$$
$$M_u = (13.075)(7.417)$$
$$\boldsymbol{M_u \approx 96.97\ \textbf{kip-ft}}$$

For a conservative check, ignore the beneficial effect of axial compression.

Using approximately:

$$d = 35.4\ \text{in}$$

the flexural reinforcement required for this relatively small moment is approximately:

$$\boldsymbol{A_{s,req} \approx 0.68\ \textbf{in}^2}$$

Provided longitudinal reinforcement:

$$\boldsymbol{A_s = 7.92\ \textbf{in}^2}$$

Therefore:

$$\mathbf{7.92 > 0.68}$$

$$\boxed{\text{PASS}}$$

The pedestal longitudinal reinforcement is therefore governed by the **minimum/detailing requirement**, not the calculated flexural demand

Step 10 - Transverse Reinforcement

The existing approved foundation drawing shows hoop/tie reinforcement around the vertical pedestal reinforcement.

For the new calculation/software implementation, I recommend:

#4 closed ties/hoops @ 10 in o.c

This provides conservative confinement around the:

$$\mathbf{12 - \#6\ vertical\ bars}$$

Use an additional tie near the top anchor/base-plate zone.

This is slightly more conservative and easier to define programmatically than relying on a fixed number of hoops.

Step 11 - Pedestal-to-Footing Development

This is **not footing design**; it is only the connection of the pedestal reinforcement into the footing.

Use:

$$\mathbf{12 - \#6\ vertical\ pedestal\ bars\ with\ standard\ hooks\ into\ footing}$$

This follows the basic configuration shown in the approved foundation drawing.

The bars must develop the full required tension force into the footing.

Required steel based purely on uplift:

$$A_{s,req} = \frac{T_u}{\phi f_y}$$

$$= \frac{103.65}{0.90(60)}$$
$$\boldsymbol{A_{s,req} = 1.92\ \mathrm{in}^2}$$

Provided:

$$A_s = 5.28\ \mathrm{in}^2$$

Therefore:

5.28 > 1.92………**PEDESTAL-TO-FOOTING STEEL TRANSFER — PASS**

The footing itself—size, thickness, punching shear, one-way shear, flexural reinforcement, soil bearing and overturning—is intentionally **outside this design**.

Anchor-to-Pedestal Interface

The updated anchor design uses:

$4 -$ 1.75" diameter anchor bolts

with:

$$h_{ef} = 40"$$

and a:

$$12" \times 12"$$

bolt arrangement inside the 39 × 39 in pedestal.

Edge distance:

$$e_x = e_y = \frac{39 - 12}{2}$$
$$\boldsymbol{e_x = e_y = 13.5"}$$

The anchor design independently checks steel tension, steel shear and concrete anchorage. For example, the calculated tension demand is 25.91 kip/bolt versus 51.3-kip steel tension capacity, while shear demand is 3.27 kip/bolt versus 30.78-kip capacity.

Therefore, the pedestal design is compatible with the complete anchor-bolt segment.